KS3 DIGITAL TECHNOLOGY

ADVANCED CONNECTIVITY: FIBRE, 5G AND SATELLITES

How data really travels: copper, fibre-optic and wireless transmission, full-fibre broadband, how mobile networks evolved from 1G to 5G and 6G, satellite internet, and the race to connect everyone.

LESSONS

1 Transmission media
2 Light in glass: full fibre
3 Mobile networks: 1G to 5G
4 5G and the road to 6G
5 Internet from space
6 Connecting everyone

LESSON 1 OF 6

TRANSMISSION MEDIA

Data travels through transmission media: electrical signals in copper cables, pulses of light in fibre-optic cables, or radio waves through the air.

STARTER · 5 MINUTES

DO NOW

List every way your data might travel between your phone and a video on a server in America.

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Compare copper cable, fibre-optic cable and wireless as transmission media

SUCCESS CRITERIA: I CAN...

- ✓ I can name three transmission media
- ✓ I can compare them for speed, range, cost and reliability
- ✓ I can choose suitable media for different situations

THREE WAYS TO TRAVEL

- Copper: electrical signals through metal wires (old phone lines, Ethernet)
- Fibre optic: pulses of light through thin strands of glass
- Wireless: radio waves through the air (Wi-Fi, 4G/5G, satellite)
- Your data usually uses all three on one journey

KEY WORDS

TRANSMISSION MEDIA	The materials or methods used to carry data, such as copper cable, fibre-optic cable or radio waves.
COPPER CABLE	A cable that carries data as electrical signals through metal wires.
FIBRE-OPTIC CABLE	A cable that carries data as pulses of light through thin strands of glass.
BANDWIDTH	The amount of data that can be sent per second, usually in megabits per second (Mbps).
LATENCY	The delay before data arrives, measured in milliseconds (ms).

TWO MEASURES OF A CONNECTION

- Bandwidth: how much data can be sent per second (megabits per second, Mbps)
- Latency: the delay before data arrives (milliseconds, ms)
- High bandwidth = fast downloads; low latency = responsive games and calls
- Different media are better at different things

CHOOSING A MEDIUM

- Fibre: huge bandwidth, long distances, but costly to install
- Copper: cheap and already everywhere, but slower over distance
- Wireless: mobile and flexible, but affected by walls, distance and interference

ACTIVITY · 15 MINUTES · PAIRS

MATCH THE MEDIUM

1. Read the six scenario cards (e.g. "a cable under the Atlantic", "a fitness tracker", "a school's main network link").
2. Choose the best transmission medium for each and give one reason.
3. Calculate how long a 1 GB file takes to download at 50 Mbps and at 900 Mbps.

In your workbook, answer:

? Which medium did you choose for "a cable under the Atlantic" and "a fitness tracker", and why?

? Show your download time calculations.

Your workbook: Lesson 1 · tick each step off as you go

QUICK CHECK

1. How does data travel through a fibre-optic cable?

A As electricity
B As radio waves
C As sound
D As pulses of light

2. What does bandwidth measure?

A The length of a cable
B The delay before data arrives
C Battery life
D How much data can be sent per second

3. What does latency measure?

A Screen size
B The delay before data arrives
C The amount of data sent
D The width of a cable

4. Which medium is NOT affected by electrical interference?

A Wi-Fi
B Fibre-optic cable
C All are affected equally
D Copper cable

QUICK CHECK: ANSWERS

1. D: As pulses of light

Fibre carries light.

2. D: How much data can be sent per second

Bandwidth = data per second.

3. B: The delay before data arrives

Latency is delay.

4. B: Fibre-optic cable

Light isn't affected by electrical noise.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain the difference between bandwidth and latency.

Look back at today's success criteria: can you tick them all?

- ✓ I can name three transmission media
- ✓ I can compare them for speed, range, cost and reliability
- ✓ I can choose suitable media for different situations

LESSON 2 OF 6

LIGHT IN GLASS: FULL FIBRE

Fibre optics send pulses of light along glass strands, trapped by total internal reflection.

STARTER · 5 MINUTES

DO NOW

How can light travel round a corner?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how fibre-optic cables carry data and describe full-fibre broadband

SUCCESS CRITERIA: I CAN...

- ✓ I can describe how light travels in an optical fibre
- ✓ I can explain the difference between FTTC and FTTP
- ✓ I can explain why full fibre is important for communities

LIGHT THAT CAN'T ESCAPE

- An optical fibre is a strand of glass thinner than a hair
- Light bounces along inside it by total internal reflection
- Data is sent as pulses of light: on = 1, off = 0
- Light in glass travels at about 200,000 km per second

KEY WORDS

TOTAL INTERNAL REFLECTION	When light hits the inside wall of a fibre at a shallow angle and bounces back in rather than escaping.
FULL FIBRE	Broadband with fibre-optic cable all the way to the building.
SUBSEA CABLE	A fibre-optic cable laid on the ocean floor to carry data between countries.

PART FIBRE OR FULL FIBRE?

- FTTC: fibre to the street cabinet, then old copper to your home
- FTTP (full fibre): fibre all the way to your home or business
- Full fibre is much faster and more reliable, and doesn't slow down over distance
- Northern Ireland has some of the best full-fibre coverage in the UK

ACTIVITY · 20 MINUTES · PAIRS

BROADBAND DETECTIVE

1. Use the broadband checker your teacher shows (e.g. Ofcom's coverage checker) to look up the school's postcode (not your home address).
2. Record what types of broadband are available and the maximum speeds.
3. Compare with one city postcode and one remote rural postcode your teacher gives you.
4. Explain the differences.

In your workbook, answer:

? Record the results for the three postcodes.

? Explain why full fibre matters for a rural farm or business.

Your workbook: Lesson 2 · tick each step off as you go

UNDER THE SEA

- Around 99% of international internet traffic travels through subsea cables
- Hundreds of cables cross the oceans, some over 10,000 km long
- They're about as thick as a garden hose in deep water
- Damage from ship anchors or sabotage is a real risk

QUICK CHECK

1. How does light stay inside an optical fibre?

A Magnetism
B Total internal reflection
C Electricity
D Gravity

2. What does FTTP mean?

A Fast transfer to phone
B Fibre all the way to the premises
C Fibre to the pole
D Fibre to the park

3. How does most international internet traffic travel?

A By satellites
B By post
C Through subsea fibre cables
D By radio

4. Why is full fibre better than FTTC for homes far from the cabinet?

A Fibre doesn't slow down over distance like copper
B Copper is faster
C Fibre uses less glass
D Distance doesn't matter for copper

QUICK CHECK: ANSWERS

1. **B: Total internal reflection**
   Light bounces off the inside walls.

2. **B: Fibre all the way to the premises**
   Full fibre to the building.

3. **C: Through subsea fibre cables**
   About 99% goes via subsea cables.

4. **A: Fibre doesn't slow down over distance like copper**
   Copper loses signal with distance.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe how data is sent through a fibre-optic cable.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe how light travels in an optical fibre
- ✓ I can explain the difference between FTTC and FTTP
- ✓ I can explain why full fibre is important for communities

LESSON 3 OF 6

MOBILE NETWORKS: 1G TO 5G

Mobile networks divide areas into cells, each served by a mast that connects phones using radio waves.

STARTER · 5 MINUTES

DO NOW

Your phone works on a moving train. How does it stay connected as you pass from town to town?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how mobile networks work and how each generation improved

SUCCESS CRITERIA: I CAN...

- ✓ I can explain how a cellular network works
- ✓ I can describe what each generation added
- ✓ I can explain the trade-off between range and speed

A NETWORK OF CELLS

- The country is divided into areas called cells
- Each cell has a mast (base station) with antennas
- Your phone talks to the nearest mast using radio waves
- As you move, it is handed over to the next mast

KEY WORDS

CELLULAR NETWORK	A mobile network made of many small areas (cells), each served by a mast.
MAST	A base station with antennas that connects mobile devices to the network.
HANDOVER	Passing a phone's connection from one mast to the next as it moves.
SPECTRUM	The range of radio frequencies used for communication.

GENERATIONS

- 1G (1980s): analogue voice calls
- 2G (1990s): digital calls and text messages
- 3G (2000s): mobile internet
- 4G (2010s): fast data, video streaming and apps
- 5G (2019 onwards): very fast, low latency, many more devices

RADIO SPECTRUM

- Mobile networks use radio frequencies called spectrum
- Low frequencies: travel far and through walls, but carry less data
- High frequencies: carry lots of data, but short range
- Governments auction spectrum to mobile companies (Ofcom in the UK)

ACTIVITY · 25 MINUTES · GROUPS OF 3

CELL PLANNER

1. Use the map of a (fictional) town and countryside your teacher gives you, with a stadium, a town centre, farms and a lough.

2. You have six masts: three low-frequency (big range, less data) and three high-frequency (small range, lots of data).

3. Place them to give everyone coverage and enough capacity for match day at the stadium.

4. Explain your choices.

In your workbook, answer:

? Where did you place the high-frequency masts and why?

? Where did you place the low-frequency masts and why?

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. What is a cell in a mobile network?

A A battery
B A prison
C A SIM card
D An area served by one mast

2. What is a handover?

A Turning off your phone
B Giving your phone to someone
C Downloading an app
D Passing your connection from one mast to the next as you move

3. Which generation introduced text messages?

A 5G
B 2G
C 4G
D 1G

4. Why do rural areas often use lower-frequency signals?

A They're newer
B They carry more data
C They travel further and through obstacles
D They're cheaper phones

QUICK CHECK: ANSWERS

1. **D: An area served by one mast**
Networks are divided into cells.

2. **D: Passing your connection from one mast to the next as you move**
It keeps calls going on the move.

3. **B: 2G**
2G brought digital calls and SMS.

4. **C: They travel further and through obstacles**
Range matters in the countryside.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe how a mobile phone stays connected while travelling on a train.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain how a cellular network works
- ✓ I can describe what each generation added
- ✓ I can explain the trade-off between range and speed

LESSON 4 OF 6

5G AND THE ROAD TO 6G

5G offers faster speeds, much lower latency and the ability to connect huge numbers of devices, enabling new uses in health, transport and industry.

STARTER · 5 MINUTES

DO NOW

A surgeon in Belfast operates a robot in a hospital in Enniskillen. What would they need from the network?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain what 5G makes possible and describe the ideas behind 6G

SUCCESS CRITERIA: I CAN...

- ✓ I can describe three improvements of 5G over 4G
- ✓ I can give examples of uses that need 5G
- ✓ I can describe what 6G might bring

WHAT 5G IMPROVES

- Speed: peak speeds many times faster than 4G
- Latency: as low as a few milliseconds in ideal conditions
- Capacity: up to a million devices per square kilometre
- Reliability for critical uses

KEY WORDS

5G	The fifth generation of mobile network, offering faster speeds, lower latency and many more connected devices.
LOW LATENCY	A very short delay before data arrives.
NETWORK SLICING	Dividing one physical 5G network into separate virtual networks with different performance.
6G	The sixth generation of mobile network, being researched and expected around 2030.

WHAT 5G MAKES POSSIBLE

- Smart factories with wireless robots
- Remote healthcare and monitoring
- Connected and self-driving vehicles talking to each other
- Smart farming: sensors on soil, livestock and machinery

THE ROAD TO 6G

- Researchers expect 6G around 2030
- Even higher frequencies and speeds
- The network itself could sense its surroundings: detecting movement, mapping rooms
- AI built in to manage the network automatically

ACTIVITY · 25 MINUTES · GROUPS OF 3

5G INNOVATION PITCH

1. Choose one area of life in Fermanagh: farming, healthcare, tourism around the lakes, or transport.
2. Design a new service that needs 5G. Explain which 5G feature it depends on: speed, low latency or many devices.
3. Identify one challenge (cost, coverage, privacy).
4. Pitch it in 60 seconds.

In your workbook, answer:

? Describe your service and the 5G feature it depends on.

? What challenge would your service face, and how could it be solved?

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. Which is a key improvement of 5G over 4G?

A Much lower latency
B Bigger phones
C It uses no radio waves
D Longer battery life always

2. About how many devices per square kilometre can 5G support?

A About 10
B About 100
C Around 1,000,000
D Exactly 4,000

3. What is network slicing?

A Turning masts off
B Cutting cables
C Sharing a pizza
D Dividing one 5G network into separate virtual networks with different performance

4. When is 6G expected?

A Around 2030
B Already everywhere
C 1990
D Never

QUICK CHECK: ANSWERS

1. **A: Much lower latency**
   Low latency is a major change.

2. **C: Around 1,000,000**
   5G supports huge numbers of devices.

3. **D: Dividing one 5G network into separate virtual networks with different performance**
   Slices give tailored service.

4. **A: Around 2030**
   It's in research.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three improvements that 5G offers over 4G.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe three improvements of 5G over 4G
- ✓ I can give examples of uses that need 5G
- ✓ I can describe what 6G might bring

LESSON 5 OF 6

INTERNET FROM SPACE

Satellite internet sends signals between a dish and satellites in space.

STARTER · 5 MINUTES

DO NOW

You're on a ship in the middle of the Atlantic. How could you get online?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how satellite internet works and compare geostationary and low Earth orbit satellites

SUCCESS CRITERIA: I CAN...

- ✓ I can describe how satellite internet works
- ✓ I can compare GEO and LEO satellites
- ✓ I can evaluate the benefits and problems of satellite constellations

HOW IT WORKS

- A dish at your home sends signals to a satellite
- The satellite relays them to a ground station connected to the internet
- Works almost anywhere with a clear view of the sky
- Used by ships, planes, remote homes and disaster zones

KEY WORDS

GEOSTATIONARY ORBIT	An orbit about 36,000 km high where a satellite stays above the same point on Earth.
LOW EARTH ORBIT	An orbit about 500-2,000 km high where satellites circle Earth quickly.
SATELLITE CONSTELLATION	A large group of satellites working together to provide a service.
SPACE DEBRIS	Old satellites, rocket parts and fragments left orbiting Earth.

HIGH OR LOW?

- Geostationary (GEO): about 36,000 km up, stays above one spot
- Signals travel so far that latency is about 600 ms round trip
- Low Earth orbit (LEO): about 500-2,000 km up, moving fast
- Much lower latency (about 20-50 ms), but needs thousands of satellites

CONSTELLATIONS AND CONCERNS

- Starlink and others have launched thousands of LEO satellites
- Brought internet to remote areas, ships and war zones
- Concerns: space debris, collisions, bright trails spoiling astronomy
- One company controlling vital connections raises questions

ACTIVITY · 20 MINUTES · PAIRS

CHOOSE A CONNECTION

1. Read the five profiles: a Fermanagh farm, a cruise ship, a research station in Antarctica, a flat in Belfast and a gamer on a Scottish island.

2. For each, recommend full fibre, 5G, GEO satellite or LEO satellite, with a reason.

3. Calculate the minimum latency for LEO using a height of 550 km (up and down, there and back).

In your workbook, answer:

? Give your recommendation and reason for the gamer on a Scottish island.

? Show your LEO latency calculation.

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. About how high is a geostationary satellite?

A About 1 km
B About 500 km
C About 36,000 km
D About 50 km

2. Why does GEO satellite internet have high latency?

A It uses copper
B Signals travel very long distances
C The satellites are slow
D Dishes are small

3. Why do LEO constellations need thousands of satellites?

A Each satellite is only overhead for a short time
B To spell words in the sky
C They are cheap
D They break every day

4. Which is a concern about large satellite constellations?

A They reduce rainfall
B They make the sea warmer
C Space debris and collisions
D They cause earthquakes

QUICK CHECK: ANSWERS

1. C: About 36,000 km

GEO is very high.

2. B: Signals travel very long distances

Distance causes delay.

3. A: Each satellite is only overhead for a short time

Fast orbits need many satellites.

4. C: Space debris and collisions

Debris is a growing problem.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Compare geostationary and low Earth orbit satellites for internet access.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe how satellite internet works
- ✓ I can compare GEO and LEO satellites
- ✓ I can evaluate the benefits and problems of satellite constellations

LESSON 6 OF 6

CONNECTING EVERYONE

The digital divide is the gap between people who have good internet access and skills and those who don't.

ADVANCED CONNECTIVITY: FIBRE, 5G AND SATELLITES

STARTER · 5 MINUTES

DO NOW

During the pandemic, schools taught online. Who might have been left out, and why?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe the digital divide and evaluate ways to connect more people

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what the digital divide is
- ✓ I can describe its causes and effects
- ✓ I can evaluate solutions to close it

WHO'S LEFT OUT?

- About a third of the world's people still don't use the internet
- Rural areas often have slower connections
- Cost of devices and data is a barrier for some families
- Skills, confidence and accessibility matter too

KEY WORDS

DIGITAL DIVIDE	The gap between people who have good internet access and skills and those who don't.
DIGITAL INCLUSION	Making sure everyone can access and use digital technology.
RESILIENCE	The ability of a network to keep working, or recover quickly, when something goes wrong.

WHY IT MATTERS

- School: homework, revision sites, online classes
- Work: job applications, remote working
- Health and money: GP appointments, banking, benefits
- People without access fall further behind

CLOSING THE GAP

- Infrastructure: full fibre, rural masts, satellites
- Affordability: social tariffs, device donation schemes
- Skills: libraries, community classes, school support
- Resilience: networks that keep working in storms and emergencies

ACTIVITY · 30 MINUTES · GROUPS OF 4

CLOSE THE GAP

1. Your group is given a budget of £10 million to reduce the digital divide in a rural county like Fermanagh.
2. Choose how to spend it across infrastructure, affordability, skills and accessibility (use the cost cards).
3. Justify your choices: who benefits, and how many people?
4. Present your plan in 90 seconds.

In your workbook, answer:

? Summarise your spending plan and the reasons for it.

? Which group of people do you think is most affected by the digital divide, and why?

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What is the digital divide?

A The gap between people with good internet access and skills and those without
B Dividing files into folders
C A broken cable
D A network split in two

2. Which is a cause of the digital divide?

A Too many websites
B Fast internet
C Free Wi-Fi everywhere
D The cost of devices and broadband

3. What is digital inclusion?

A Adding cables
B Including more apps on a phone
C Making sure everyone can use digital technology
D Joining a social network

4. What does network resilience mean?

A Keeping working or recovering quickly when something goes wrong
B Being cheap
C Being very fast
D Using only one cable

QUICK CHECK: ANSWERS

1. **A: The gap between people with good internet access and skills and those without**
   It's about unequal access.

2. **D: The cost of devices and broadband**
   Cost is a major barrier.

3. **C: Making sure everyone can use digital technology**
   Inclusion closes the gap.

4. **A: Keeping working or recovering quickly when something goes wrong**
   Resilient networks have backups.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three causes of the digital divide.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what the digital divide is
- ✓ I can describe its causes and effects
- ✓ I can evaluate solutions to close it

END OF TOPIC

ADVANCED CONNECTIVITY: FIBRE, 5G AND SATELLITES

Revise each lesson in your workbook: the notes, key words and quick checks. Then take the topic test when Mr Butler sets it.